Exercise 9: Trim and Linearization

a) Give a set of states that suffices to fully describe the rigid body motion of an aircraft. Classify the states as either “longitudinal” or “lateral”. Explain (by giving an example) why your particular choice of states is not the only possible solution.

b) State the primary control effectors of an aircraft. State the primary effect of each control (force or moment)?

c) Transform the well-known nonlinear differential equation of the spring-mass-damper system

$$m \cdot l^2 \cdot \ddot{\varphi}(t) = -m \cdot g \cdot l \cdot sin\, \varphi(t) - F_d\big(\dot{\varphi}(t)\big) + F(t) \cdot l \cdot \cos \varphi(t)$$

with the nonlinear damping force $F_d\big(\dot{\varphi}(t)\big) = d \cdot \dot{\varphi}(t) \cdot \cos\frac{\varphi(t)}{\dot{\varphi}_{max}}$ to a linear system. Linearize around the steady-state operating point $\varphi(0) = \varphi_0$.

d) Linearize the equation for the y-position (which is decoupled from the longitudinal motion) and the equation for the bank angle around the condition: „steady-state straight flight without sideslip “.

$$0 = \dot{y} - V \cdot \sin\chi \cdot \cos\gamma = \dot{y} - V \cdot \sin(\Psi + \beta) \cdot \cos(\,\Theta - \alpha)$$

$$0 = \dot{\phi} - p - q \cdot \sin\Phi \cdot \tan\Theta - r \cdot \cos\Phi \cdot \tan\Theta$$

Additional data:

Density at MSL according to ISA standard atmosphere:	$\rho_{s,MSL} = 1{,}225\,\frac{kg}{m^3}$
Pressure at MSL according to ISA standard atmosphere:	$p_{s,MSL} = 1{,}01325e5\ Pa$
Temperature at MSL according to ISA standard atmosphere:	$T_{s,MSL} = 288{,}15\ K$
Gravitational acceleration:	$g_0 = 9{,}80665\,\frac{m}{s^2}$
Specific gas constant:	$R = 287{,}05\,\frac{J}{kg\,K}$
Polytropic exponent:	$n = 1{,}235$
Isentropic exponent:	$\kappa = 1{,}4$
Radius of the earth:	$r_{earth} = 6356766\ \mathrm{m}$